

Market Mood Radar

A directional commodity-sentiment scoring platform

PREPARED	LIVE APPLICATION	STATUS	BUILD SCOPE
17 June 2026	marketresearch-tau.vercel.app	Complete & deployed	14-step roadmap, finished

1 Executive summary

Market Mood Radar scores six commodities — Gold, Silver, Platinum, WTI Crude, Natural Gas and Copper — on a single explainable **directional “mood” scale from –100 to +100**, plus an overall market mood. Each score fuses five independent signals (price technicals, macro factors, news tone, broad sentiment and a geopolitical oil/gas overlay) into one number with a plain-English explanation of *why*.

The application is fully built, tested and live. Its data refreshes daily with no manual intervention, it surfaces its own reasoning, and it has been **backtested and validated against two years of historical price data**. Headline honesty point: the price-driven model shows a **modest but real edge — about 52% directional accuracy versus a ~50% baseline**. It is a research signal, **not financial advice**.

2 What the application does

- Scores 6 commodities on a 5-band scale (Strong Bull → Cautious Optimism → Neutral/Mixed → Risk-Off → Strong Bear).
- Explains every score with the signals that drove it, and lets users drill from a reason down to the underlying news headlines.
- Tracks how individual scores and the overall mood move over time.
- Lets users set score-threshold alerts (email delivery pending domain setup).
- Backtests the price-driven model and reports its historical accuracy per commodity.
- Documents its own scoring methodology in an in-app page.

3 Tools & technologies used

Framework & language

TOOL	ROLE
Next.js 16 (App Router, Turbopack)	Full-stack React framework — pages and API routes
React 19	User-interface rendering
TypeScript 5 (strict)	Type-safe codebase
Node.js	Runtime for server code and tooling scripts

UI & visualisation

TOOL	ROLE
Tailwind CSS v4 + shadcn/ui	Styling and component library
Recharts	Charts: price, RSI, MACD, mood-over-time, backtest
next-themes	Light / dark theme switching

Data sources

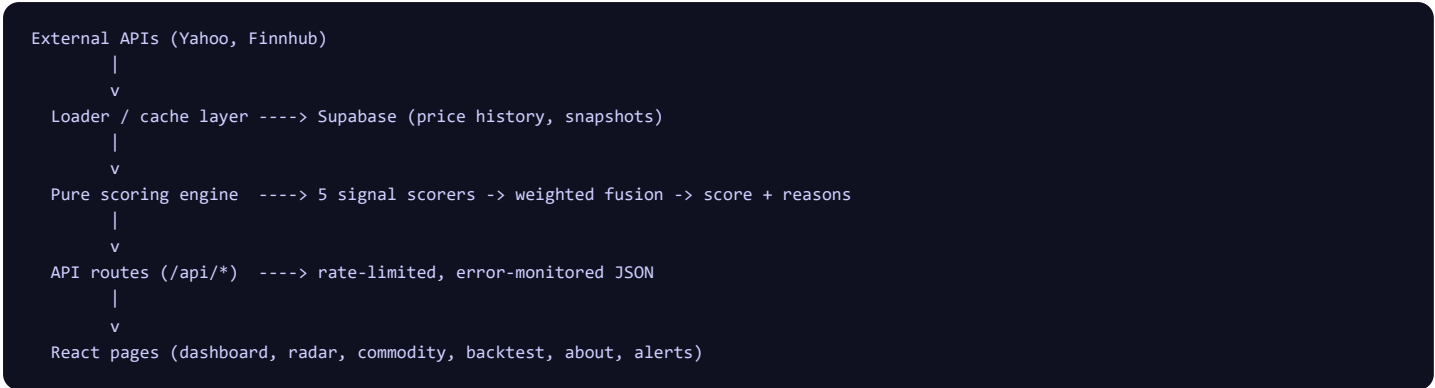
SOURCE	PROVIDES	NOTES
Yahoo Finance	Daily prices (commodities + DXY, VIX, 10Y, S&P)	No key; unofficial; fallback host built in
Finnhub	News headlines + economic calendar	Free tier (calendar is paid / unavailable)

Storage, operations & quality

TOOL	ROLE
Supabase (PostgreSQL)	Caches 2 years of prices; stores score snapshots and alerts
Resend	Alert email delivery (pending sender-domain verification)
Vercel + cron-job.org	Hosting, deployment, and the scheduled daily/hourly data refresh
Sentry	Production error monitoring
Jest + Playwright	Automated unit tests and end-to-end browser tests

4 How it works — architecture

The system is built around a **pure scoring engine** that is fully deterministic and testable: it takes pre-fetched data in and returns scores out, with no network calls of its own. Data fetching, caching and side effects live in separate layers.



Scoring method, in plain terms: each signal returns a direction (−1...+1), a confidence (0–1) and reasons. The engine blends them by *weight × confidence*, ignores neutral signals, and damps low-conviction reads (fewer than ~3 agreeing signals are scaled down). The result is scaled to −100...+100 and labelled. The two price-driven weights were tuned by the backtest optimiser (see section 5).

5 Validation & backtesting

A core design principle was **proving the signal rather than asserting it**. A backtesting engine replays the price-driven score across ~2 years of history and compares each day’s score to the actual forward price move at 1, 3 and 7 days. A weight optimiser then grid-searches the price-signal weights, picking the best on a training window and **only adopting it if it still beats the defaults on a held-out, more-recent test window** — an explicit guard against over-fitting.

50.2%

Default weights — held-out hit rate

52.4%

Optimised weights — held-out hit rate

+2.26pp

Out-of-sample improvement (n ≈ 2,100)

The optimiser settled on emphasising trend (moving averages) and macro context. This is a **statistically meaningful but modest** edge — appropriately so for a price-only model on liquid markets. News and geopolitical signals are live-only (not stored historically), so they are deliberately excluded from the backtest.

6 Features & pages

PAGE	PURPOSE
Dashboard	Commodity grid: score, direction, confidence, sparkline
Radar	Mood gauge, “what’s moving”, explainability with headline drill-down, correlation matrix, mood-over-time chart, news & calendar
Commodity detail	Price + Bollinger / RSI / MACD charts, indicator readings, score history
Backtest	Per-commodity hit-rate stats, score history, returns-by-band
About	Methodology, signal weights, backtest proof, honest limitations
Alerts	Score-threshold alerts (email pending)

Cross-cutting: light/dark theme, “not financial advice” footer, mobile-responsive layout, and graceful empty/error/stale states throughout.

7 Reliability, security & operations

- **Automatic data refresh** — a scheduled job tops up prices and records score snapshots (hourly via cron-job.org, daily via Vercel). The app stays current with no manual reset.
- **Rate limiting** — public API routes are capped (default 60 requests/min per IP) with standard headers and 429 responses.
- **Error monitoring** — server errors are reported to Sentry (verified working); a no-op when unconfigured, so it adds no risk.
- **Graceful degradation** — every external source fails soft: a missing key or 403 yields an annotated empty result, never an error page.
- **Resilient price source** — price requests fall back to a second Yahoo host on failure.
- **Secured refresh endpoint** — protected by a secret token.

8 Testing

LAYER	TOOL	COUNT	COVERS
Unit	Jest	114 tests	Indicators, signal scorers, fusion engine, backtest math, weight optimiser, rate limiter, error-payload builders
End-to-end	Playwright (Chromium)	7 tests	Every main page renders without error; theme toggle works

Both suites pass and the production build is clean (0 type errors).

9 Honest limitations

Not financial advice. Market Mood Radar is a directional research signal with a modest backtested edge — for information only, not a recommendation to buy or sell any asset.

- The economic-calendar signal is unavailable on the current (free) Finnhub tier.
- Alert emails are not yet sending (Resend sender domain not verified).
- Price data comes from an unofficial Yahoo API — free but not guaranteed.
- The backtest covers the price model only; news/geopolitical weights are not historically testable and are set by judgement.
- Model weights do not re-optimize automatically (intentional — avoids chasing noise); refreshing them is a deliberate developer step.

10 Build completion

All six planned phases / fourteen steps are complete: a test safety-net, relevance fixes (broad sentiment, headline de-duplication), stronger signals (two-year price cache), trust features (backtesting and a methodology page), UX upgrades (trend charts, headline drill-down, theme toggle) and hardening (rate limiting, error monitoring, end-to-end tests). One optional feature, the alerts upgrade, is deferred pending email setup.

Companion documents in the project: README.md (specification & scoring methodology), HANDOVER-CLAUDE.md (full engineering build log), NOTE.md (management take-over checklist). — Market Mood Radar, 2026.